

Program: Diploma in Food Processing Technology	
Course Code: 6421B	Course Title: Dairy Processing Technology
Semester: 6	Credits: 4
Course Category: Program elective	
Periods per week: 4(L:4 T:0 P: 0)	Periods per semester: 60

Course Objectives:

1. To understand the properties and characteristics of milk.
2. To study the importance and effect of pasteurization, homogenization, centrifugation, and membrane separation in processing of milk
3. To impart knowledge on preparation methods of various dairy products

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of milk composition	Dairy processing Technology

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the properties, reception and storage of milk in dairy industries.	15	Understanding
CO2	Illustrate the principle, importance and various methods of sterilization and pasteurization	15	Understanding
CO3	Describe the effect of homogenization, importance of centrifugation and membrane separation in dairy industry	15	Understanding
CO4	Explain the various methods of butter manufacturing, ice-cream production, yogurt manufacturing, and cheese theory, including the types of freezers and cheese.	15	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the properties, reception and storage of milk in dairy industries		
M1.01	Classify the type of Milk and its composition	5	Understanding
M1.02	Identify the physical, Chemical and Thermal Properties	5	Understanding
M1.03	Discuss the reception, storage and cooling of milk	5	Understanding
Contents: Milk-Types-Composition-factors affecting composition of milk, Physical-Chemical and Thermal Properties of milk, Reception and storage-cooling of milk.			
CO2	Illustrate the principle, importance and various methods of sterilization and pasteurization		
M2.01	Outline the pasteurization techniques	3	Understanding
M2.02	Discuss about LTLT/holding pasteurization	4	Understanding
M2.03	Discuss about HTST pasteurization	4	Understanding
M2.04	Explain about sterilization, UHT processing, vacreation	3	Understanding
	Series Test –I	1	
Contents: Pasteurization - principles, objectives and methods, LTLT/holding pasteurization-types, advantages and disadvantages, HTST pasteurization-Principle, theory and function of HTST pasteurizer, advantages and disadvantages. Sterilization, UHT processing, Vacreation			

CO3	Describe the effect of homogenization, importance of centrifugation and membrane separation in dairy industry		
M3.01	Explain the Homogenization	5	Understanding
M3.02	Explain the importance of centrifugation in milk processing	5	Understanding
M3.03	Explain the importance of membrane separation processes in milk processing-	5	Understanding
Contents: Homogenization theory, effect of homogenization of milk, Importance of centrifugation in milk processing-advantages and disadvantages, Importance of membrane separation processes in milk processing-ultrafiltration-advantages and disadvantages			
CO4	Explain the various methods of butter manufacturing, ice-cream production, yogurt manufacturing, and cheese theory, including the types of freezers and cheese.		
M4.01	Explain the butter manufacturing process	4	Understanding
M4.02	Describe the ice-cream manufacturing process	3	Understanding
M4.03	Explain the yogurt manufacturing process	3	Understanding
M4.05	Interpret the cheese manufacturing process	4	Understanding
	Series Test–II	1	
Contents: Butter manufacturing methods, Ice-cream manufacture-overrun-types of freezers used, Yogurt manufacturing process, Cheese-theory, types			

Text / Reference

T/R	Book Title/Author
R1	Tufail Ahmed, "Dairy Plant Engineering and Management", CBS Publishers and Distributors, New Delhi, 2001, ISBN-13: 978-8122501186
R2	Pieter Walstra, Jan T M Wouters, Tom J Geurts, "Dairy Science and Technology", Taylor Francis Group, CRC press. 2006. ISBN-13: 978-0824727635
R3	De Sukumar, "Outlines Of Dairy Technology", Oxford University Press.2015, ISBN-13: 978- 0195611946
R4	Spreer, Edgar "Milk and Dairy Product Technology". Marcel Dekker, 2005. ISBN-13: 978-0824700942

R5	Selia, Jane dos Reis Coimbra and Jose A. Teixeir “Engineering Aspects of Milk and Dairy Products”. Jane Selia dos Reis Coimbra & Jose A. Teixeir, CRC Press, 2009, ISBN13: 978-1420090222
R6	Eiri Board “Handbook of Milk Processing Dairy Products and Packaging Technology” Engineers India Research Institute, 2008. ISBN-13: 978-8186732960

Online Resources

Sl.No	Website Link
1	https://dairynutrition.ca/en/milk-quality/homogenization/why-milk-homogenized-and-what-are-its-effects
2	https://khatabook.com/blog/how-to-manufacture-butter-step-by-step-process/
3	https://thecheeseweb.com/7-types-of-cheese
4	https://www.sciencedirect.com/science/article/pii/S0022030279831975
5	https://dairyprocessinghandbook.tetrapak.com/chapter/centrifugal-separators-and-milk-standardization